

In this paper, we observe that typical manifolds,

We promise that: equivalent relation on X generated by: condition means the smallest equivalent relation on X satisfying the condition.

Rigorously, $\sim := \bigcap R_1 \times R_2$
 $R_1 \times R_2 \in \mathcal{X} \times \mathcal{X}$ is eq-rel
 $R_1 \times R_2$ satisfies the condition

And, $I = [0, 1]$, the subspace of the euclidean space $\mathbb{R}$.

$$\begin{array}{ccc} X & & \\ \pi \downarrow & \searrow f & \\ X/\sim & \xrightarrow{\tilde{f}} & Y \end{array}$$

Cylinder

Def.

Define an equivalence relation $\sim$ on $I \times I$ generated by: $\forall y \in I, (0, y) \sim (1, y)$
The quotient space $I^2/\sim$ is called the Cylinder.

Prop. The cylinder $I^2/\sim$ is Homeomorphic to $S^1 \times I \subseteq \mathbb{C} \times \mathbb{R}$.

Proof. Define a map

$$f: I^2 \rightarrow S^1 \times I: (x, y) \mapsto (e^{2\pi i x}, y)$$

$$\begin{array}{ccc} I^2 & & f \\ & \searrow & \\ \pi \downarrow & & S^1 \times I \\ I^2/\sim & \xrightarrow{\tilde{f}} & \end{array}$$

Then, f is continuous onto closed map, because

1. Continuous

$(x, y) \mapsto e^{2\pi i x}$ and $(x, y) \mapsto y$ are continuous, so f is also continuous.

2. Onto

Given $(e^{i\theta}, b) \in S^1 \times I$ where $\theta \in [0, 2\pi]$, take $x = \frac{\theta}{2\pi}$ and $y = b$, then clear.

3. Closed

Given closed set $C \subseteq I^2$, C is compact since I^2 is compact.

So, the continuous image $f[C]$ is also compact in $S^1 \times I$,

therefore $f[C]$ is closed, by $S^1 \times I$ is Hausdorff.

Moreover, $f(x_1, y_1) = f(x_2, y_2) \Leftrightarrow e^{2\pi i x_1} = e^{2\pi i x_2}$ and $y_1 = y_2$
($\Leftrightarrow e^{2\pi i(x_1 - x_2)} = 1$)

$$\Leftrightarrow \left(x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\} \right) \text{ and } y_1 = y_2$$

$$\Leftrightarrow (x_1, y_1) \sim (x_2, y_2)$$

Therefore, f induces a Homeomorphism

$$\tilde{f}: I^2/\sim \rightarrow S^1 \times I: [(x, y)] \mapsto f(x, y).$$

It is injective and well-defined by $(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow f(x_1, y_1) = f(x_2, y_2)$

And surjective is given by the f is surjective.

Finally, the continuity is given by: $\tilde{f} \circ \pi = f$ is continuous, and,

given closed set $C \subseteq I^2/\sim$, $\pi^{-1}[C]$ is closed, so the ontoness of π gives

$\tilde{f}[C] = \tilde{f}[\pi[\pi^{-1}[C]]] = f[\pi^{-1}[C]]$ is closed, since f is closed map. $\square$

Torus

Def.

Define an equivalence relation $\sim$ on $I \times I$ generated by:

$$\begin{cases} \forall x, y \in I, (x, 0) \sim (x, 1) \text{ and } (0, y) \sim (1, y) \\ (0, 0) \sim (1, 0) \sim (0, 1) \sim (1, 1) \end{cases}$$

The quotient space $I^2/\sim$ is called a Torus.

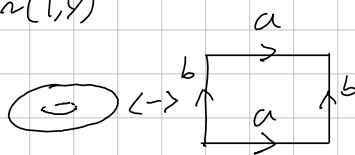
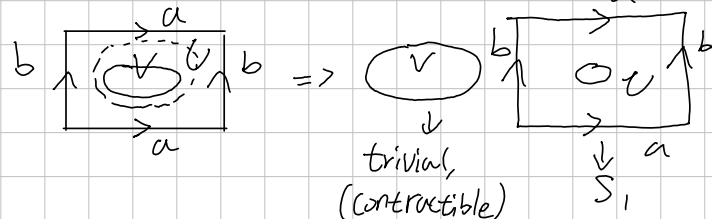
Prop. The Torus $I^2/\sim$ is Homeomorphic to $S^1 \times S^1 \subseteq \mathbb{C}$.

Proof. Define a map

$$f: I^2 \rightarrow S^1 \times S^1: (x, y) \mapsto (e^{2\pi i x}, e^{2\pi i y})$$

Then, f is continuous onto closed map,

Fundamental group of the Torus,



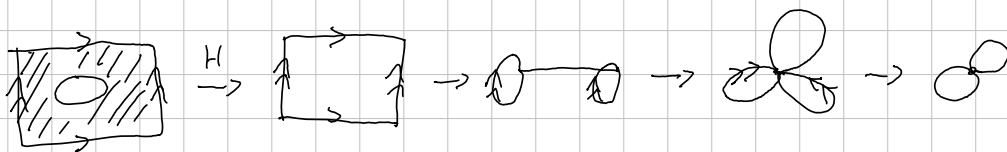
$$\langle a, b \mid aba^{-1}b^{-1} = e \rangle$$

$$\cong \mathbb{Z} \times \mathbb{Z}$$

$I^2/\sim \cong S^1 \times S^1$ gives

$$\pi_1(I^2/\sim) \cong \pi_1(S^1 \times S^1)$$

$$\cong \pi_1(S^1) \times \pi_1(S^1)$$



Therefore, $\pi_1(U) * \pi_1(V) / \pi_1(U \cap V)$

$$\cong \langle a \rangle * \langle b \rangle / \langle aba^{-1}b^{-1} \rangle$$

$$\cong \mathbb{Z} \times \mathbb{Z}.$$

Projective space.

Def. Define an equivalence relation on $\mathbb{R}^{n+1} \setminus \{0\}$ generated by:

$$x \sim y \Leftrightarrow \exists t \in \mathbb{R} \text{ such that } tx = y.$$

The quotient space $\mathbb{RP}^n \stackrel{\text{def}}{=} (\mathbb{R}^{n+1} \setminus \{0\}) / \sim$ is called a Real Projective space.

Prop. $\mathbb{RP}^1$ is Homeomorphic to the circle S^1

Proof. Define a map

$$f: \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow S^1: (x,y) \mapsto \frac{1}{x^2+y^2} \cdot (x^2-y^2, 2xy)$$

It is derived by $r \cdot e^{i\theta} \mapsto e^{i \cdot 2\theta}$, so f is onto. Moreover, it is continuous map, being rational function, entrywise.

And, observe that

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow \theta_1 - \theta_2 = k\pi \text{ for some } k \in \mathbb{Z} \text{ where } e^{i\theta_j} = (x_j, y_j) \\ \Leftrightarrow e^{i \cdot 2(\theta_1 - \theta_2)} = 1 \quad (j=1,2).$$

$$\Leftrightarrow f(x_1, y_1) = f(x_2, y_2).$$

Therefore, the induced map

$$\tilde{f}: \mathbb{RP}^1 \rightarrow S^1: [(x,y)] \rightarrow f(x,y)$$

is well-defined, injective, surjective.

Moreover, for $\tilde{\iota}: S^1 \rightarrow \mathbb{R}^2 \setminus \{(0,0)\}: (x,y) \mapsto (x,y)$,

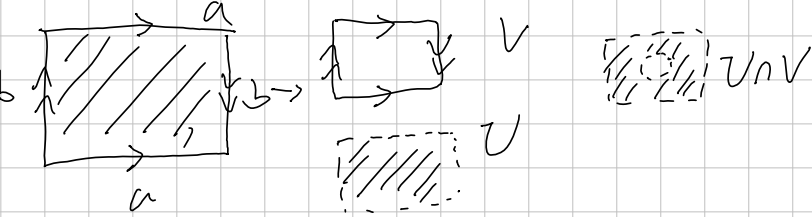
$$\pi \circ \tilde{\iota}: S^1 \rightarrow \mathbb{RP}^1$$

is continuous onto map, and since S^1 is compact, so $\mathbb{RP}^1$ is compact. Finally, $\mathbb{RP}^1$ is compact and S^1 is Hausdorff, thus $\tilde{f}$ is closed map, therefore $\tilde{f}$ is also closed map, so $\mathbb{RP}^1 \stackrel{\text{def}}{=} S^1 \quad \square$

$$\begin{array}{ccc} \mathbb{R}^2 \setminus \{(0,0)\} & & \\ \pi \downarrow & \searrow f & \\ \mathbb{RP}^1 & \xrightarrow{\tilde{f}} & S^1 \\ & f = \tilde{f} \circ \pi & \end{array}$$

Covering space of $\mathbb{R}P^n$.

Klein bottle.



$$\rightarrow \pi_1(V) = \langle a \rangle * \langle b \rangle$$

$$\pi_1(U) = \{1\}$$

$$\pi_1(U \cap V) = \langle aba^{-1}b \rangle$$

$$\text{so, } \pi_1(K) \cong \langle a \rangle * \langle b \rangle / \langle aba^{-1}b \rangle$$

$$= \langle a, b \mid aba^{-1}b = 1 \rangle$$

Möbius band.

Sphere

Thm. $D^n/S^{n-1} \cong S^n$

Proof. Define a map

$$f: D^n \rightarrow S^n: X = (x_1, \dots, x_n) \mapsto (2x_1 \sqrt{1 - \|x\|^2}, 2x_2 \sqrt{1 - \|x\|^2}, \dots, 2x_n \sqrt{1 - \|x\|^2}, 2(\|x\|^2 - 1))$$

